A Lidar Benchmark for Desert Rover Traversability: Slip Face Detection, Hazard Layers and Route Planning from Coarse Elevation Models on Active Sand Dunes

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***Abstract*—For most deserts the only free elevation models a small wheeled rover can plan on have a posting of about 30 m, while the main hazard on active dunes, the slip face, is often only metres long. We built a benchmark on six North American dune fields with one metre airborne lidar as truth. Ten families of hazard layers, from slope and interpolation to boosted trees, a convolutional network, learned super resolution and imagery fusion, were scored against a lidar face map and used by a slip aware planner at operating points chosen on the other five sites. At 10 m the best layer raises mission success from 17 % to 64 %; at 30 m no computable layer adds more than 5 points to 8 %, even for faces longer than 60 m, while a face map oracle reaches 81 %, and fewer routes still are climbable on the lidar. In post hoc closed loop runs with line of sight sensing, 30 m priors raise success from 30 % to at most 48 %, and not clearly when endpoints are chosen from the coarse map. Code, results and the prediction log are public.**

***Index Terms*—field robotics, traversability, airborne lidar, digital elevation models, sand dunes, slip faces, hazard mapping, path planning, benchmark**

I. Introduction

Small wheeled ground robots are proposed for survey, inspection and search in sand deserts, including the dune seas of the Arabian Peninsula. Before such a robot drives, its route is planned on a prior map, and for most deserts the only free digital elevation models (DEMs) are global products with a posting of about 30 m [1], [2], [3]. The dominant hazard on active dunes is the slip face, the lee side of a dune held near the angle of repose of dry sand, about 30 to 34 degrees [4], where wheel slip rises until a small vehicle can no longer climb [5], [6]. Many slip faces are shorter than 30 m (Section IV-B), so a 30 m cell can hold one while its mean slope is gentle (Fig. 1).

That a coarse grid understates slope is long known [7]. What is not known is how much this matters for a planner, which hazard layer a practitioner should compute from a coarse map, how fine a map must be, and whether imagery, a different product or onboard sensing changes the answer. This paper answers these questions with a benchmark on six North American dune fields where one metre airborne lidar provides the truth.

The contributions are (1) a benchmark that scores hazard layers against a lidar face map whose meaning is the same at every resolution, on surfaces from 2 to 60 m, real elevation products and with imagery; (2) a comparison of ten families of hazard layers, learned ones trained leave one site out; (3) a planning evaluation at held out operating points, with degraded face maps, vehicle feasibility and a closed loop with line of sight sensing; and (4) an analysis of detection against face length and grid spacing.

II. Related Work

Overhead priors. Airborne lidar and overhead imagery at metre scale have served as route priors for ground vehicles [8], [9]. Drones can map ahead of the robot [10] or learn terrain properties from the air [11]. Planetary missions plan from orbital data [12], [13]. These priors are at metre scale; we ask what a coarse prior, the only free kind for most deserts, hides.

Slip on sand. Geometric costmaps score slope and roughness on a grid [14], [15], [16]. On loose sand slip rises steeply with slope [5], [6], [17], [18]. Slip can be learned from appearance [19], desert vehicles have planned on overhead and onboard maps [20], and planners propagate slip uncertainty into routes [21], [22], [23], but they assume that the map resolves the slope, which fails on dunes at 30 m.

Dunes in elevation models. Coarser grids understate slopes [7]. Dune morphometry from DEMs and imagery is well developed [24], and deep learning maps crestlines from DEM derivatives [25]. Global 30 m products have been ranked against lidar [26]; the radar mission TanDEM-X, the source of the Copernicus GLO-30 DEM, is weaker over sand [27], and open DEM error in a dune sea grows above 15 degrees [28]. Learned super resolution can sharpen coarse DEMs [29]. We know of no benchmark that scores coarse map hazard layers and planners for slip faces against lidar truth across resolutions.

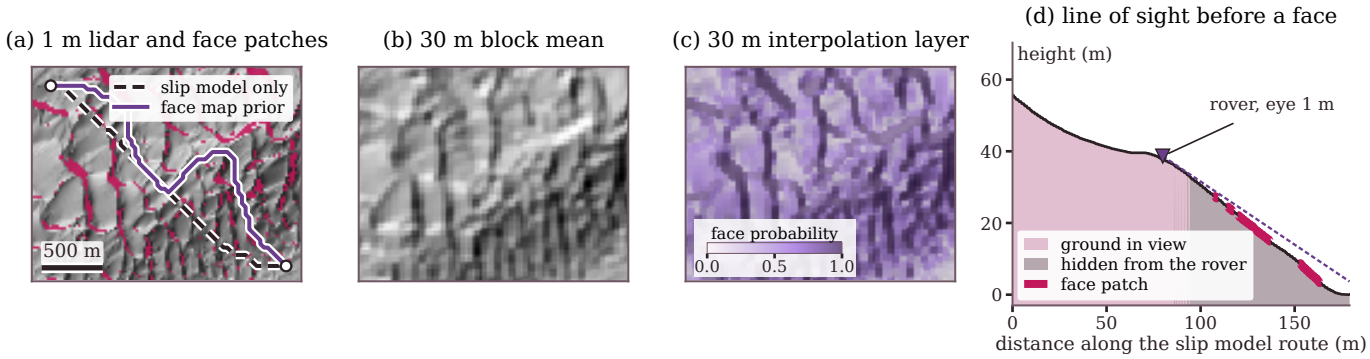


Fig. 1. What a 30 m map hides, at Great Sand Dunes. (a) One metre lidar hillshade with face patches (rose) and the 30 m routes of one E1 pair: the slip model alone (dashed) crosses faces that a face map prior (solid) avoids. (b) The 30 m block mean of the same ground, the map the planner sees. (c) The calibrated interpolation layer computed from it. North is up. (d) Lidar profile along the slip model route (vertical scale exaggerated): from an eye 1 m above the ground, the faces ahead lie in ground hidden behind the convex brink.

TABLE I
LIDAR WINDOWS (USGS 3DEP ONE METRE DEM; FLOWN: ACQUISITION PERIOD OF THE LIDAR PROJECT), THE NUMBER OF FACE PATCHES AND THE SHARE OF CELLS HOLDING A FACE PATCH POST AT 10 AND 30 M.

Site	Lat., lon.	km^2	Relief	Flown	Patches	10 m	30 m
Dumont Dunes, CA	35.65, -116.27	14.4	322 m	2024 to 25	4693	36 %	49 %
White Sands, NM	32.82, -106.29	6.3	20 m	2020	789	15 %	40 %
Kelso Dunes, CA	34.91, -115.72	9.7	240 m	2024 to 25	2184	15 %	38 %
Great Sand Dunes, CO	37.75, -105.54	14.8	246 m	2019 to 20	2065	16 %	32 %
Algodones Dunes, CA	32.92, -115.10	20.0	122 m	2021 to 23	3131	10 %	27 %
Little Sahara, UT	39.70, -112.39	6.4	99 m	2020 to 21	1776	11 %	35 %

III. BENCHMARK

A. Sites, truth and label

Six windows of one metre bare earth lidar from the 3D Elevation Program (3DEP) of the U.S. Geological Survey (USGS) [30] cover active dune fields (Table I), chosen from 20 m slope images. A post is *steep* when its Horn slope [31] at 1 m is 28° or more, a little below the angle of repose of dry sand [4]; voids never count as steep. A *face patch* is an 8 connected group of at least 10 steep posts (10 m^2). Face patches are steep ground, not verified slip faces. For a coarse cell c with footprint F_c the label is the face map

$$y_c = \begin{cases} 1 & \text{if some } x \in F_c \text{ lies in a face patch,} \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

which a planner needs; its meaning does not change with the cell size, although its prevalence does (Table I).

B. Inputs

Idealised: block means of the lidar at 2, 5, 10, 15, 20, 30 and 60 m (I10 and I30 in the tables) (no product noise or time gap). *Real:* the USGS 1/3 arc second DEM (R10, about 8×10 m) [32], NASADEM (N30, February 2000) [33] and Copernicus GLO-30 (G30, 2011 to 2015) [1], each scored on its own geographic grid, with labels from the lidar posts inside each native cell; no horizontal shift is applied. For planning, GLO-30 is interpolated to the 30 m lidar grid. USGS source footprints show that R10 was made from the truth lidar at all six sites, so it tests the product chain only. *Imagery:* one Sentinel-2 level 2A (surface reflectance) scene per site [34] (10 m; lidar year, cloud below 5 %, lowest sun, chosen by rule) and National Agriculture Imagery Program (NAIP) images [35] (0.3 to 0.6 m, resampled to 1 m).

C. Hazard layers

The layers (supplement Table S2) are slope, ruggedness [36], plane roughness as in GESTALT [16], an aspect rule, interpolation (the steepest slope after cubic spline interpolation to 1 m), learned super resolution (a dilated network [37] that predicts a 5 m residual), a logistic subgrid detector trained only on synthetic dune profiles, which ships with the planner, gradient boosted trees [38], [39], [40] on 14 terrain features, a fully convolutional network [41] trained with Adam [42], and boosting on terrain plus Sentinel-2 or NAIP features. Every layer trained on lidar is trained leave one site out [43], and a layer that is not a probability is calibrated by logistic regression [44] on the other five sites when a planner needs one.

D. Planner

The planner searches the 8 connected grid of the input. For an edge with along track grade $\tan\alpha$ (uphill positive) and cross slope $\tan\beta$,

$$d = \sqrt{(\tan\alpha + f_r)^2 + \tan^2\beta}, \qquad i = -i_0 \ln\left(1 - d/\tan\phi\right) \tag{2}$$

give the traction demand and the slip ($i = \infty$ when $d \geq \tan\phi$), with friction angle ϕ, motion resistance coefficient f_r and slip scale i_0; the slip law is the exponential shear relation of Janosi and Hanamoto [45], [46] written for the ratio of demand to capacity. Edge costs use the nominal parameters: an edge is passable when $i \leq i_{\max}$ and costs $\ell/(1-i)$ for length ℓ. For the classes, 128 parameter sets are drawn; a cell is passable for a draw when at least four of its eight headings are, q_c is the share of draws for which it is, and a cell is BLOCKED when $q_c \leq 0.05$, TRAVERSABLE when $q_c \geq 0.95$ and its face probability $p_c < \tau$, and UNKNOWN otherwise. The

main vehicle, the *posterior vehicle*, takes ϕ, f_r and i_0 from a posterior of the exponential family fitted by Markov chain Monte Carlo to twelve published slip curves of other wheeled vehicles (Mars Science Laboratory test rovers in Mars Yard sand and at Dumont Dunes [5], Scarab and El Dorado on tilted sand [47], Spirit on Husband Hill [48]; 5 to 95 % ranges ϕ 30.6 to 43.7°, f_r 0.03 to 0.32, i_0 0.33 to 1.21) and $i_{\max}$ from an assumed range of 0.4 to 0.7 (nominal values 36.9°, 0.13, 0.63 and 0.5; nominal climbing limit 15.9°). A vehicle with narrower assumed ranges (limit 25.5°) is the sensitivity. Three policies share the graph: *slip only*, *gate* (TRAVERSABLE cells only) and *weight* (no BLOCKED cell, cost times $1 + w\,p_c$ of the cell entered); gate and weight refuse descents of 28° or more. Every layer is run at $\tau \in \{0.02, \ldots, 0.9\}$ and $w \in \{1, \ldots, 100\}$ and reported at the setting that maximises mission success on the other five sites.

E. Scoring and closed loop

One hundred pairs per site, at least 1 km apart, are drawn twice: E1 from cells whose lidar posts are all gentle (truth used) and E2 from cells whose coarse slope is below 15° (no truth). A route, the polyline through its cell centres sampled every 0.5 m on the lidar, is *unsafe* when a sample falls on a face patch, and a *mission success* when it exists and is not unsafe. In the closed loop (20 pairs per site) the rover follows the plan on a 5 m grid that inherits the coarse classes and face probabilities ($w = 5$), with slip costs from the coarse surface interpolated to 5 m; at every step it senses the face patches within $R = 20$ m, blocks those cells, frees the others and replans when its route is blocked, with up to 300 replans.

F. Statistics and prediction log

Detection is scored by the area under the receiver operating characteristic curve (AUC); AUC intervals come from a spatial block bootstrap [49] with 300 m blocks; route outcomes carry two stage bootstrap intervals (sites, then pairs). With six sites, an exact sign flip test over sites can reach at best $p = 1/32$, and $k/32$ after Holm correction [50] over k comparisons, so no site level comparison here can be significant at 0.05; we report per site counts and intervals instead and make no site level significance claim. The design of every result in Tables III and IV was committed to a prediction log before it ran, with numeric predictions for 43 of them, but after earlier results on the same sites were known; the checks in Section IV-E are post hoc. The predictions of stages 25 to 28 (31 hits, 12 misses) and three amendments made before the affected results were known are in the supplement.

IV. Results

A. Which layer sees faces

Table II gives the mean AUC of every layer for the face map label. On 10 m block means most layers rank face cells well (interpolation 0.97, boosting 0.97, slope 0.90), and R10 behaves like the 10 m block mean (interpolation 0.96). At 30 m every elevation layer loses skill: with default settings interpolation is best (0.82), ahead of boosting (0.81) and the network (0.80), and the detector (0.74) is above only slope, NAIP features and the aspect rule. Super resolution (0.84), the network trained four times longer (0.84) and boosting with five more features (0.83) do better; nested tuning changes nothing (0.81). The 10 m data still hold the information: the largest of the nine 10 m interpolation scores in a 30 m cell reaches 0.96.

TABLE II

Detection of the face map label of Eq. (1): mean AUC over the six sites. I: lidar block means; I10→30: the largest of the nine 10 m scores in each 30 m cell; R10: USGS 1/3 arc second DEM, resampled by USGS from the same lidar at all six sites; N30: NASADEM; G30: GLO-30; R10, N30 and G30 on their own grids, where the network and super resolution were not run (on GLO-30 interpolated to the 30 m lidar grid they reach 0.72 and 0.70). *Trained on the lidar of the other five sites. ‡Post hoc, pixel level. Best per column in bold (ties at two decimals all bold); · not run.

Layer	I10	I30	I10→30	R10	N30	G30
Slope	0.90	0.69	·	0.90	0.62	0.65
Ruggedness	0.94	0.76	·	0.93	**0.63**	0.69
Roughness	0.80	0.77	·	·	·	·
Aspect rule	0.83	0.61	·	·	·	·
Interpolation	**0.97**	0.82	**0.96**	**0.96**	0.62	0.70
Super resolution*	·	0.84	·	·	·	·
Detector	0.95	0.74	0.95	0.95	0.56	0.65
Boosting*	**0.97**	0.81	**0.96**	0.95	**0.63**	**0.72**
Boosting+*	**0.97**	0.83	·	·	·	·
Network*	0.96	0.80	0.95	·	·	·
Network, 6000 steps*	·	0.84	·	·	·	·
Sentinel-2 only*	0.68	0.74	·	·	·	·
Boosting + Sentinel-2*	**0.97**	**0.85**	·	·	·	·
NAIP only*	0.62	0.67	·	·	·	·
Boosting + NAIP*	**0.97**	0.84	·	·	·	·

Imagery adds a little. Boosting on terrain plus Sentinel-2 features reaches 0.85 at 30 m (+0.04 over terrain alone), and with NAIP 0.84; imagery alone is weaker (0.74 and 0.67), and at 10 m it adds nothing (0.97). Super resolution and boosting with Sentinel-2 score above interpolation at all six sites and the detector below it at all six (supplement Table S4), which with six sites is the strongest evidence such a design can give.

The real 30 m products are worse than the 30 m block mean: on their own grids the best layer reaches 0.72 on GLO-30 and 0.63 on NASADEM, which predates the lidar by about two decades; interpolating GLO-30 to the lidar grid changes little (best 0.72). Shifting GLO-30 by the offset that best aligns it with the lidar raises interpolation only from 0.70 to 0.70 (post hoc), so misregistration does not explain the loss; dune migration, the radar's weakness over sand [27] and the product's own smoothing remain, and we cannot separate them.

The label matters for small faces. With the earlier cell label (at least 10 steep posts) the 30 m interpolation scores 0.82; requiring 100 or 300 steep posts raises it to 0.89 and 0.97, so a 30 m map ranks large faces well and small ones poorly; a 30 degree threshold changes little (0.82), nor do two rules that join fragmented patches (post hoc: 0.82 with hysteresis, 0.82 with merging), and against faces of at least 60 m only interpolation reaches 0.88.

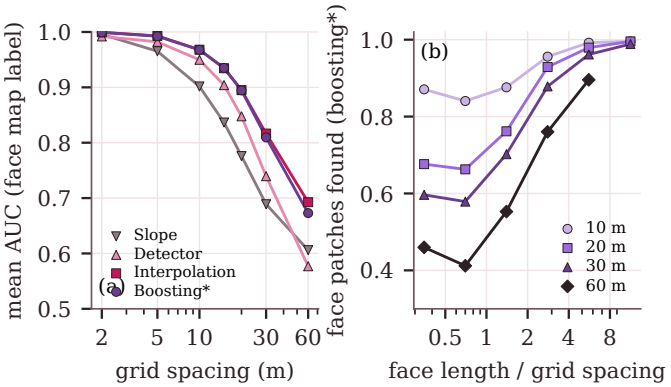


Fig. 2. Resolution and scale (lidar block means, face map label). (a) Mean AUC over six sites against the grid spacing. (b) Share of face patches found by boosting (a cell holding the patch is flagged at the threshold that flags 10 % of face free cells) against patch length over grid spacing, in bins with edges 0.5, 1, 2, 4 and 8, for four spacings; bins with fewer than 50 patches are omitted. Patch length: major axis of the ellipse with the same second moments. *Trained leave one site out.

B. How fine a map must be

Fig. 2a shows how detection falls with grid spacing. Interpolation scores 1.00 at 2 m, 0.97 at 10 m, 0.82 at 30 m and 0.69 at 60 m. Fig. 2b relates the fall to face size. At every spacing, boosting finds nearly all faces longer than four grid cells (96 % of faces 4 to 8 cells long at 30 m, 90 % at 60 m), but at 30 and 60 m far fewer faces shorter than half a cell (60 % at 30 m, 46 % at 60 m). Longer patches touch more cells and so have more chances to be flagged, which accounts for part of this rise. By count most patches are short: 75 % are shorter than one 30 m cell, but they hold only 7 % of the face posts, and with a hysteresis rule that joins fragments of one face the shares are 50 % and 3 % (post hoc).

C. Planning

Table III gives mission success at the operating points chosen on the other five sites (every weight: supplement S4). With the posterior vehicle the gate routes at most 34 % of pairs at 10 m and 42 % at 30 m, whatever the layer, because the slip model is uncertain on most dune slopes (most refusals are at an endpoint: 76 % for boosting at 30 m and $\tau = 0.2$). The weight policy routes almost every pair and is chosen for every layer; on 10 m maps it mostly takes the largest weight of the grid ($w = 100$), but with weights up to 1000 (post hoc) held out success changes by at most a few points (interpolation 66 % at 10 m, 13 % at 30 m). At 30 m slope and the network fall below the slip model alone: they flag much of the dune, so the penalised routes grow longer and still meet faces.

On 10 m maps every layer helps: mission success rises from 17 % with the slip model alone to 64 % with interpolation (+47 points, two stage interval +28 to +64, positive at all six sites), 59 % with boosting and 50 % with the detector. On 30 m maps the gain is small: interpolation adds +5 points (interval +2 to +9) and boosting with Sentinel-2 +5 (interval +1 to +9) to 8 %, and no layer computed from GLO-30 has an interval above zero. With the assumed vehicle the 30 m gain of interpolation

TABLE III

MISSION SUCCESS (%) AT THE OPERATING POINT CHOSEN ON THE OTHER FIVE SITES (GATE THRESHOLD OR WEIGHT), 100 PAIRS PER SITE, POSTERIOR VEHICLE. E1: ENDPOINTS CHOSEN WITH THE TRUTH; E2: CHOSEN FROM THE COARSE MAP ONLY. †TWO STAGE BOOTSTRAP INTERVAL OF THE GAIN OVER THE SLIP MODEL ENTIRELY ABOVE ZERO (NOT CORRECTED FOR THE NUMBER OF CELLS). · NOT RUN. *TRAINED LEAVE ONE SITE OUT.

	I10		I30		G30
Layer	E1	E2	E1	E2	E1
Slip model only	17	9	8	3	6
Slope	35†	28†	6	3	6
Interpolation	64†	53†	13†	5	7
Detector	50†	33†	7	4	5
Boosting*	59†	47†	9	4	7
Network*	51†	40†	6	3	7
Boosting + Sentinel-2*	56†	46†	13†	7†	·
Face map (oracle)	84†	80†	81†	68†	80†

is +7 points. With E2 endpoints the slip model alone reaches 3 % at 30 m and the gain of interpolation is not clearly above zero (interval 0 to +6).

The face map oracle reaches 81 % on the 30 m grid. The registered degraded face maps (supplement Table S6, weight policy at $w = 5$) show how good that information must be: the oracle gives 55 % there, displaced by 10 m 43 % and by 30 m 11 %, and maps built with the 28 degree rule from simulated 2 m and 5 m surveys give only 23 % and 8 % because they miss faces. The outcome is robust to the scoring details (supplement S5).

D. Onboard sensing

Table IV adds onboard sensing within 20 m. A slip face lies behind a crest, out of sight of a rover on the windward slope (Fig. 1d), so the main runs, decided after the registered ones, sense a 5 m cell only when it is visible from an eye 1 m above the ground. With the slip model alone as 30 m prior the rover then succeeds on 30 % of the E1 pairs, and 65 % touch a face. Interpolation, boosting with Sentinel-2 and super resolution raise this to 48 %, 48 % and 48 % (interpolation +18 points, interval +6 to +33, higher at all six sites), and the oracle to 74 %. With E2 endpoints no computable prior has an interval above zero (best boosting, 38 % against 29 %). If each face cell is detectable with probability 0.8, success falls to 12 % with the slip model prior and 68 % with the oracle. No pair used up its replan budget. The registered idealised sensor that sees every face (last column) gives 80 % to 85 % whatever the prior and overstates what sensing can do.

E. Post hoc checks

These checks were decided after the main results were known and are post hoc. *What the vehicle can drive.* Mission success counts only face contact. When every 5 m step of a route on the 5 m lidar surface must also be climbable by the nominal posterior vehicle, with no descent of 28° or more, vehicle success on 10 m maps is 47 % with interpolation against 9 % for the slip model alone. On 30 m maps no computable layer exceeds 9 %, and the face map oracle reaches

TABLE IV

CLOSED LOOP WITH ONBOARD SENSING WITHIN 20 M, 20 PAIRS PER SITE, POSTERIOR VEHICLE, WEIGHT $w = 5$: SUCCESS (%; GOAL REACHED WITHOUT TOUCHING A FACE). LINE OF SIGHT: A 5 M CELL IS SENSED ONLY WHEN VISIBLE FROM AN EYE 1 M ABOVE THE GROUND; $p_d = 0.8$: EACH FACE CELL IS DETECTABLE WITH PROBABILITY 0.8. LAST COLUMN: THE REGISTERED IDEALISED SENSOR THAT SEES EVERY FACE WITHIN THE RADIUS. †TWO STAGE BOOTSTRAP INTERVAL OF THE GAIN OVER THE SLIP MODEL PRIOR ENTIRELY ABOVE ZERO. · NOT RUN. *TRAINED LEAVE ONE SITE OUT. LINE OF SIGHT RUNS ARE POST HOC.

	Line of sight				Ideal
	I30			I10	I30
Prior layer	E1	E2	E1, p_d 0.8	E1	E1
Slip model only	30	29	12	41	80
Detector	34	35	·	·	85
Interpolation	48†	36	·	63†	82
Boosting*	42†	38	21†	·	84
Boosting + Sentinel-2*	48†	36	·	·	·
Super resolution*	48†	36	·	·	·
Network, 6000 steps*	38	34	·	·	·
Face map (oracle)	74†	64†	68†	82†	85

only 17 %: 21 % of its successes are drivable, because its routes go round faces over slopes the coarse map shows as gentle.

Face size. Scored against faces of at least 60 m only, 30 m interpolation adds +8 points over the slip model alone (29 % to 37 %), against +42 on 10 m maps, so the gap is not a matter of small patches.

Other checks (supplement S6). On 5 m block means interpolation reaches 86 %; at 30 m super resolution and the longer trained network add +5 and +2 points.

V. DISCUSSION AND LIMITATIONS

What the results mean. A 30 m elevation model keeps enough of the dune shape to rank face cells, but not enough to plan around faces: one missed face fails a route of a kilometre, so recall, not ranking, decides planning, and the gap to 10 m maps remains for long faces only. A 30 m map also hides the steep climbs between faces, so even the face map oracle drives few routes the vehicle can climb. With line of sight a coarse prior helps, by +12 to +18 points with E1 endpoints but not clearly with E2 endpoints, while a perfect face map more than doubles success. The useful prior is fine face and slope information placed within a few metres.

Limitations. The truth is steep ground: rock, ruts and vegetated mounds also exceed 28 degrees, three windows are off road vehicle areas, and the audit of 600 patches is not finished, so label precision is unknown. We chose the windows, had used them before, and keep the prediction log in our own repository; the line of sight, vehicle and label checks are post hoc. The vehicle is a posterior fitted to slip curves of other vehicles, and no slip model family transferred between published studies in our validation (median held out error 0.20 to 0.24), so absolute outcomes are uncertain. No rover was driven, the sensor model has no range noise, the 10 m results use lidar block means (most deserts have no free 10 m product), and six sites give no power for site level tests. Arabian fields are untested: a megadune face 100 m high spans about five 30 m cells and should be visible, smaller flank faces would not, and dunes migrate.

VI. CONCLUSION

A 30 m elevation model ranks slip faces better than chance but cannot keep a small rover off them. On six lidar dune fields no hazard layer computable from such a map, including imagery fusion and learned super resolution, added more than a few points of mission success at its held out operating point, while the same kind of layer on a 10 m map raised it from 17 % to 64 %; the gap remained for long faces only and widened when the vehicle also had to climb every step. With line of sight sensing a 30 m prior helped by about +12 to +18 points, and a face map prior far more, so fine, well placed face information is what a rover needs. Next are the label audit, Arabian fields with a drone survey, and a rover with measured slip on the same sand.

DATA AND CODE

USGS 3DEP lidar and DEMs, NASADEM and NAIP are in the public domain; Sentinel-2 data are provided under the Copernicus free licence; Copernicus GLO-30 is used under its licence (produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA; all rights reserved). Code, results, the prediction log and the scripts that regenerate every number and figure are at https://github.com/hudasol/dune-rover (stages 25 to 30).

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